Gene regulation of Hedgehog interacting protein (HHIP) in Chronic Obstructive Pulmonary Disease

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Introduction

Chronic obstructive pulmonary disease (COPD) causes 3 million people death and is estimated to become the third cause of mortality by 2030 (WHO., 2020). COPD is a progressive, irreversible and disabling disorder that causes severe respiratory insufficiency.

The clinical manifestations and genomic profiles and heterogeneous and change across individuals, challenging its research.

Chang et al. made the characterization and subclassification of the COPD patients using network- based stratification or NBS, in which consider the interaction networks between genes and gene expression in blood and they found new subtypes of COPD.

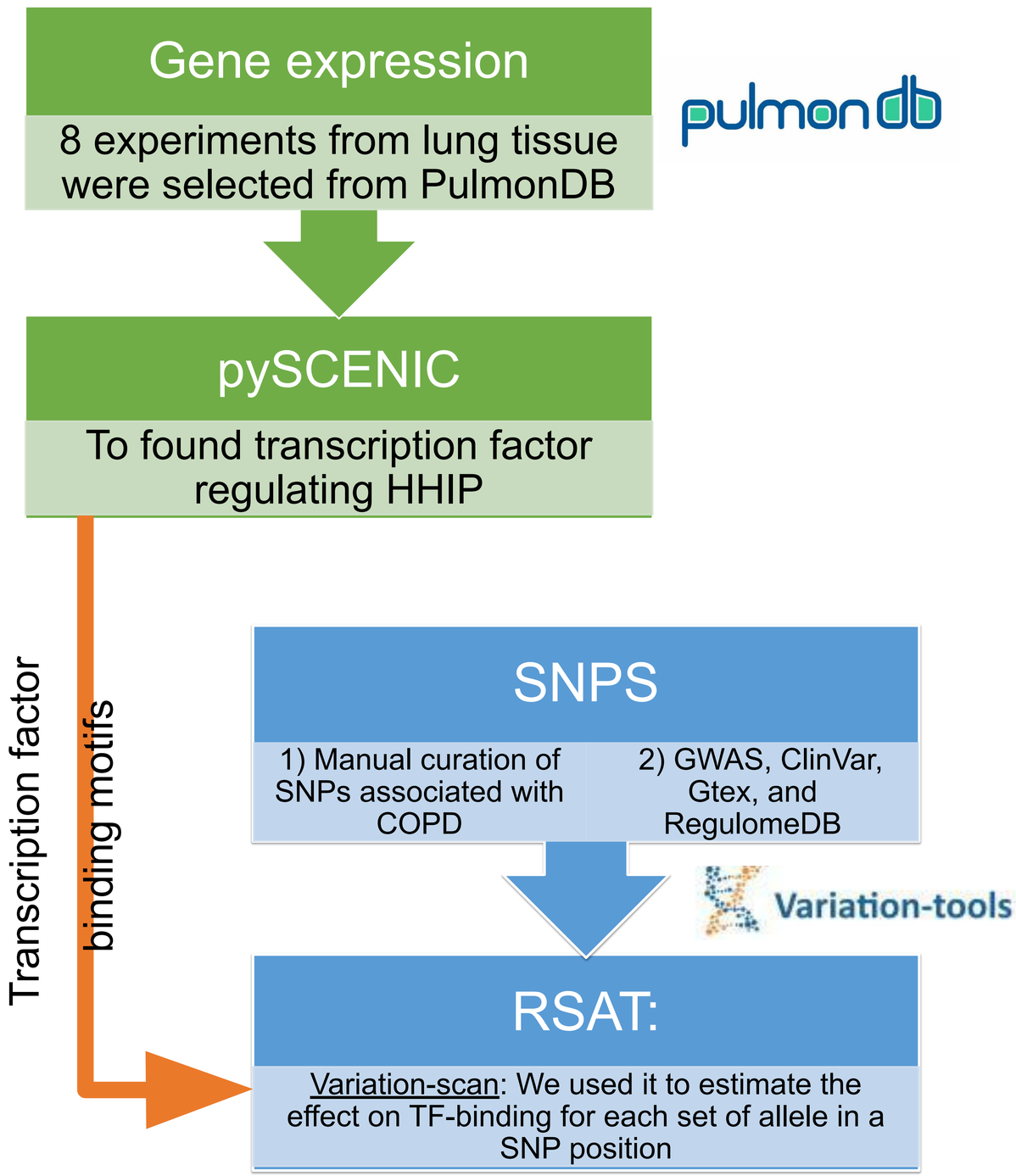
One of the pathways that regulates lung morphogenesis is the Hedgehog pathway (Hh). Sonic Hedgehog (SHH) is the most expressed ortholog as a ligand in the HH pathway. SHH pathway also includes PTCH1 as a transmembrane receptor specific of SHH, when it is activated, promote transcriptional factors like GLI2 and GLI3 that leads to the activation of target genes like hedgehog interacting protein (HHIP).

Objective

We aim to understand if COPD associated variants in the regulatory region of HHIP affect the motif binding sites of transcription factors that regulate HHIP expression.

Materials and methods

We first performed manual curation of papers reporting HHIP gene expression, in a direct (e.i. molecular studies) or indirect (e.i. transcriptomic high-throughput studies) approach. Then, we used PulmonDB to analyze the expression of all Hh pathway components in COPD, and calculate regulons using pyscenic. Finally we searched for genetic variants that may be affecting regulation of HHIP using VariationScan.



Results

Gene expression of HHIP and HH pathway components in COPD experiments from PulmonDB

-Idiopathic pulmonary fibrosis (IPF) is considered an antagonist chronic disease of COPD

Figure 1. A) Gene expression of HHIP in data from PulmonDB, p-values are nominal. B)Gene expression of HH pathway components and other genes associated in PulmonDB.

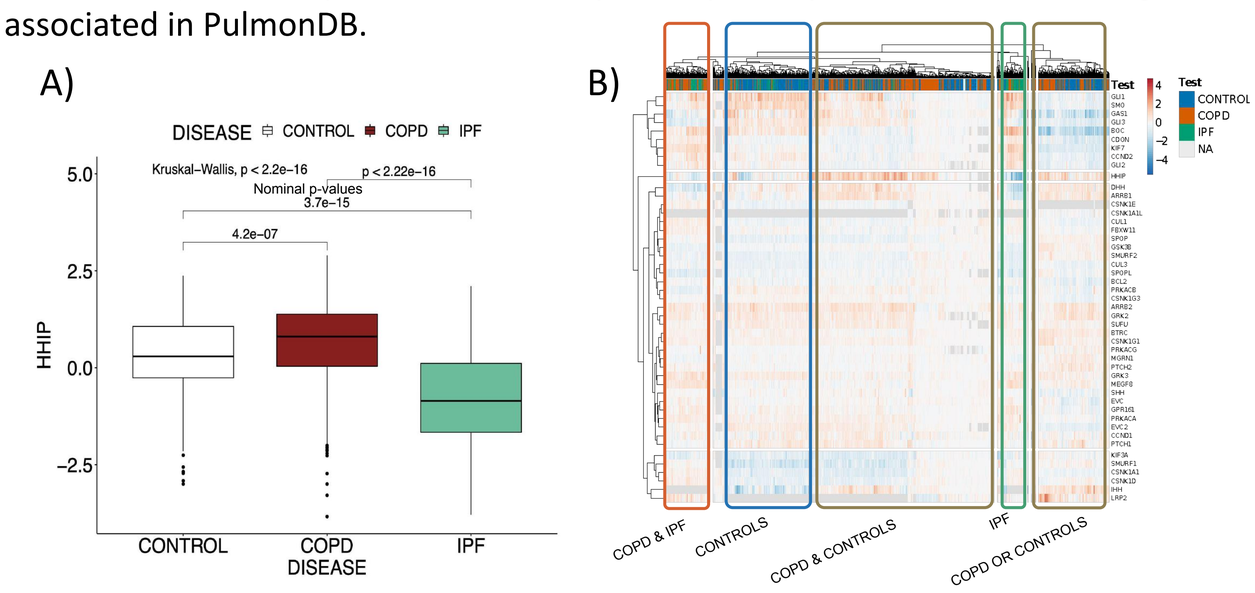


Figure 2. HHIP and its possible regulators. A) Transcription factors regulating 293 regulons that included HHIP. B) Regulon from pySCENIC, in which HLF is the potential transcription factor of different co-expressed genes including HHIP.

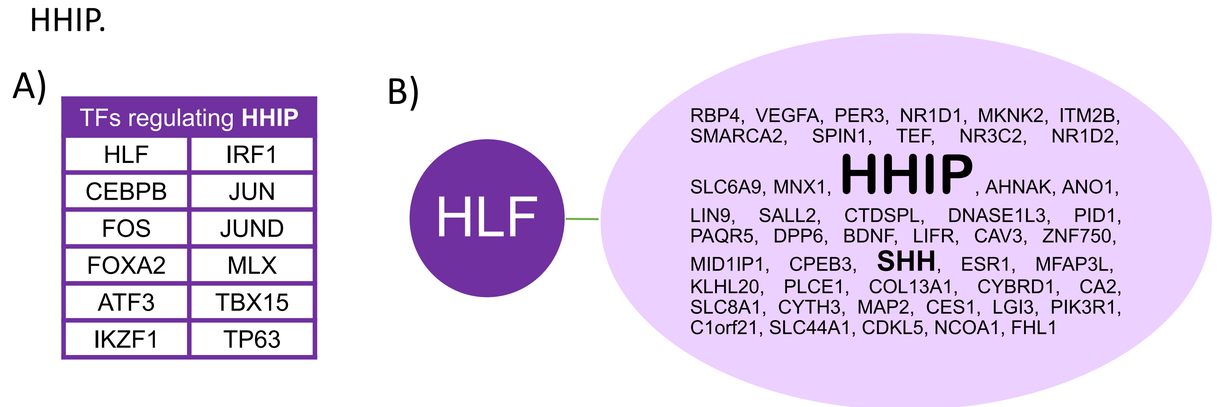


Figure 3. HHIP and SNPs associated with COPD. A) Genomic region of HHIP in human genome using UCSC browser. SNPs in databases were taken from GTEx, Clinvar, GWAS, and RegulomeDB.

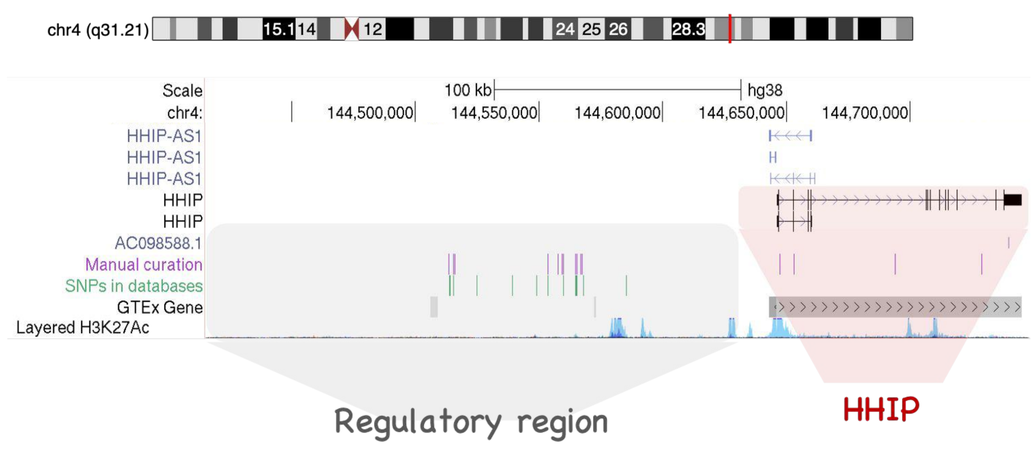
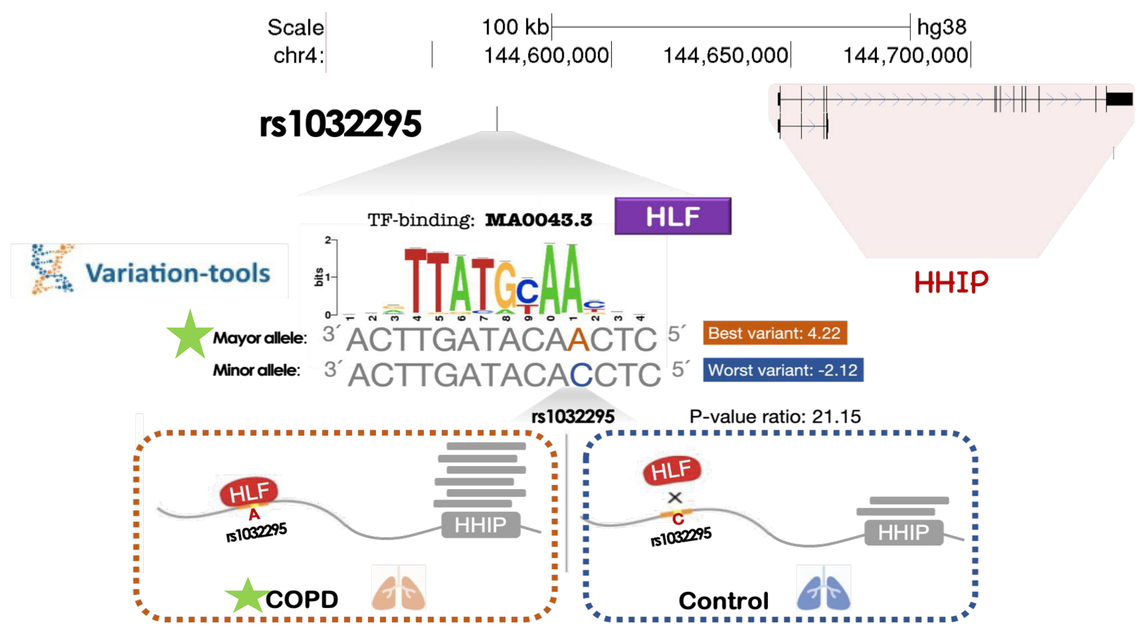


Figure 4. TF-motif of HLF. Results from VariationScan showed affinity lost for HLF motif (MA0043.3) when the minor allele C for rs1032295 is present (Allele A has been associated with COPD).



Conclusions

- Variants associated with COPD in the regulatory region of HHIP may has an impact in transcriptional binding motifs
- The mayor allele for rs1032295 changes the affinity of HLF in the promoter region of HHIP

Acknowledgments

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- LAVIS
- Laboratorio Internacional de Investigación sobre el Genoma Humano (LIIGH) UNAM

References

-*WHO | Burden of COPD*. Who.int. (2020). Retrieved 26 April 2020, from https://www.who.int/respiratory/copd/burden/en/.
-Villaseñor-Altamirano, A. B., Moretto, M., Maldonado, M., Zayas-Del Moral, A., Munguía-Reyes, A., Romero, Y., ... & Selman, M. (2020). PulmonDB: a curated lung disease gene expression database. *Scientific reports*, *10*(1), 1-9.
-Santana-Garcia, W., Rocha-Acevedo, M., Ramirez-Navarro, L., Mbouamboua, Y., Thieffry, D., Thomas-Chollier, M., Contreras-Moreira, B., van Helden, J., Medina-Rivera, A., 2019. RSAT variation-tools: An accessible and flexible framework to predict the impact of regulatory variants on transcription factor binding. Comput. Struct. Biotechnol. J. 17, 1415–1428.
-Vrahatis, A. G., Dimitrakopoulos, G. N., Tasoulis, S. K., Georgakopoulos, S. V., & Plagianakos, V. P. (2019, December). Single-cell regulatory network inference and clustering from high-dimensional sequencing data. In *2019 IEEE International Conference on Big Data (Big Data)* (pp. 2782-2789). IEEE.
-Chang, Y., Glass, K., Liu, Y. Y., Silverman, E. K., Crapo, J. D., Tal-Singer, R., ... & Castaldi, P. (2016). COPD subtypes identified by network-based clustering of blood gene expression. *Genomics*, *107*(2-3), 51-58
-Kugler, M. C., Joyner, A. L., Loomis, C. A., & Munger, J. S. (2015). Sonic hedgehog signaling in the lung. From development to disease. *American journal of respiratory cell and molecular biology*, *52*(1), 1-13
-Wang, B., Zhou, H., Yang, J., Xiao, J., Liang, B., Li, D., ... & Yu, H. (2013). Association of HHIP polymorphisms with COPD and COPD-related phenotypes in a Chinese Han population. *Gene*, *531*(1), 101-105.
-Evaluation of HHIP Polymorphisms And Their Relationship With Chronic Obstructive Pulmonary Disease Phenotypes. *International Journal of Chronic Obstructive Pulmonary Disease*, *14*, 2267.
-Zhou, X., Baron, R. M., Hardin, M., Cho, M. H., Zielinski, J., Hawrylkiewicz, I., ... & Thibault, D. (2012). Identification of a chronic obstructive pulmonary disease genetic determinant that regulates HHIP. *Human molecular genetics*, *21*(6), 1325-1335
-Obeidat, M. E., Hao, K., Bossé, Y., Nickle, D. C., Nie, Y., Postma, D. S., ... & Elliott, W. M. (2015). Molecular mechanisms underlying variations in lung function: a systems genetics analysis. *The Lancet Respiratory Medicine*, *3*(10), 782-795.
-Fawcett, K. A., Obeidat, M., Melbourne, C. A., Shrine, N., Guyatt, A. L., John, C., ... & Weiss, S. (2019). SNPs associated with HHIP expression have differential effects on lung function in males and females. *bioRxiv*, 594457.
-Zhou, X., Qiu, W., Sathirapongsasuti, J. F., Cho, M. H., Mancini, J. D., Lao, T., ... & Lomas, D. A. (2013). Gene expression analysis uncovers novel hedgehog interacting protein (HHIP) effects in human bronchial epithelial cells. *Genomics*, *101*(5), 263-272.
Bártholo, T. P., Porto, L. C., Pozzan, R., Nascimento, A., & Da Costa, C. H. (2019).